2022-2023 (SPRING) YTU CIVIL ENG. MECHANICS DIV.

STRENGTH of MATERIALS II

MIDTERM EXAM *26.04.2023*

Name:
Group:
No:
L=
Sign:

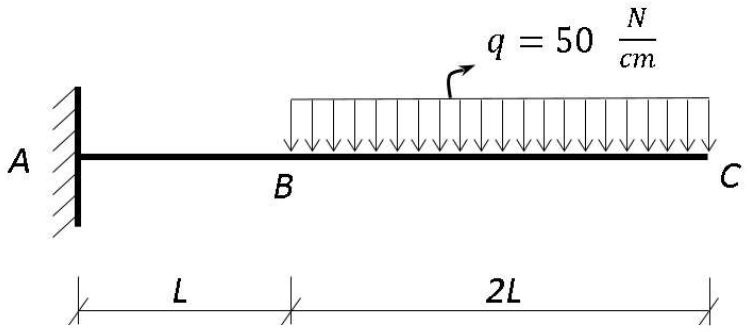
Explanation: Parameter L
Number a: is the last digit of your student number.
Number b: is the second last digit of your student number.

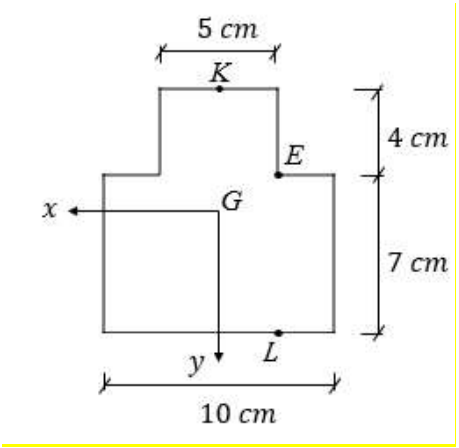
Example for **"L=1ab (cm)":** Student number: xxxxxx94, a=4, b=9 then **L=149 cm**

Duration: **30 minutes** (solution + uploading) **Good luck**

2) (34p) Draw the normal and shear stress distributions for the most critical cross-section of the cantilever beam shown in the figure.

Check the section of the beam at the neck point ***E*** using the **Maximum Shear Stress Hypothesis** (Tresca hypothesis). $\left(\sigma_{all} = 80\ MPa\right)$





(Cross-section)